

Venkata Sai Deepak Kuppuru

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Data professional with **3+ years of experience** delivering data-analytics and predictive-modeling solutions for **fintech and healthcare**. Adept at turning complex datasets into actionable insights that **strengthen fraud-detection, reduce operational costs, and improve patient outcomes** using **Python, SQL, and BI tools**.

Work Experience

Paypal | Data Scientist

May 2024 - Aug 2025

- Consolidated and cleaned **10M+ payment-transaction records** from five data sources, improving data reliability and cutting manual prep time by **40%** for downstream fraud-risk analysis.
- Performed **exploratory and statistical analysis** that identified **3 high-risk merchant categories** and **revealed 25% seasonal spikes** in fraudulent activity across multiple global regions.
- Partnered with the **fraud-risk team to define KPIs and business rules**, improving alert prioritization and reducing false positives by **18%**, saving an estimated **\$2 M annually** in investigation costs.
- Designed **interactive Tableau dashboards** delivering real-time visibility to risk managers, enabling them to act on alerts **50% faster** and reduce average investigation time from **4 hours to 2 hours**.
- Delivered **monthly and ad-hoc executive reports** summarizing fraud trends and KPIs, helping leadership implement policies that **cut chargeback losses by 12% year-over-year**.

Optum | Data Scientist

Oct 2022 -Jul 2023

- Integrated and cleaned **1.2 M+ longitudinal EHR and claims records** from multiple hospital systems, **improving data completeness by 38%** to support population-health analytics.
- Conducted **in-depth exploratory analysis and cohort segmentation** (age, chronic-condition clusters, SDOH factors) that uncovered a **22% higher readmission risk** in unmanaged diabetic patients.
- Developed and tuned **predictive models** (logistic-regression & gradient-boosting) to flag high-risk members for early care-management, achieving an **82% F1-score** and improving care-team triage efficiency by **27%**.
- Built **HIPAA-compliant automated data pipelines and validation checks** that cut manual reporting time by **45%** and reduced data-quality issues by **30%** across analytics workflows.
- Designed **interactive Power BI dashboards** for clinicians and operations leaders to track population-level risk, care-gap closure, and 30-day readmission KPIs in **near-real-time**, improving decision-making speed.
- Collaborated with **clinical, actuarial, and compliance teams** to turn model insights into targeted care-coordination programs, achieving an **estimated 8% drop** in avoidable admissions within two quarters.

Optum | Data Analyst

Feb 2022 - Sep 2022

- Collected, cleaned, and standardized **250K+ patient claims and visit records**, reducing missing data by **35%** and improving dataset reliability for analysis and downstream predictive modeling efforts.
- Performed **exploratory data analysis** on 12-month claims and demographics to uncover **15% higher readmission rates** in chronic-care cohorts, driving targeted cost-reduction initiatives.
- Built a **logistic-regression model** to predict high-risk patients using age, previous visits, and chronic-condition history, achieving **78% accuracy** and enabling proactive patient-care intervention strategies.
- Designed **interactive Tableau dashboards** visualizing utilization trends, cost by service type, and predicted readmission risk, enabling data-driven care decisions.
- Implemented automated weekly and monthly **SQL/Excel reports** for utilization and cost KPIs, cutting manual reporting effort by 50% and improving data accuracy for leadership reviews.
- Delivered **self-serve user-journey datasets** and reporting layers, powering analytics and experimentation that drove a **30% lift in conversion rates** and remarketing effectiveness.

SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn, SciPy, Seaborn, Matplotlib, Keras, PyTorch), R, SQL, C

Data Engineering & Databases: ETL, Hadoop, Hive, Apache Spark, MySQL, MongoDB

Machine Learning & AI: Statistical Analysis, Causal Inference, A/B testing, Attribution methods, Hypothesis testing

Tools & Technologies: Tableau, Power BI, Azure, AWS, Linux, Docker, Kubernetes, Git, Excel, CI/CD Pipeline.

Modeling: Machine Learning (Clustering, Classification, Recommendation), Deep Learning, NLP, Statistical Modeling.

EDUCATION

Master of Science, Computer Science – University of Illinois Springfield

Bachelor of Technology, Computer Science specialization in Artificial Intelligence – KL University